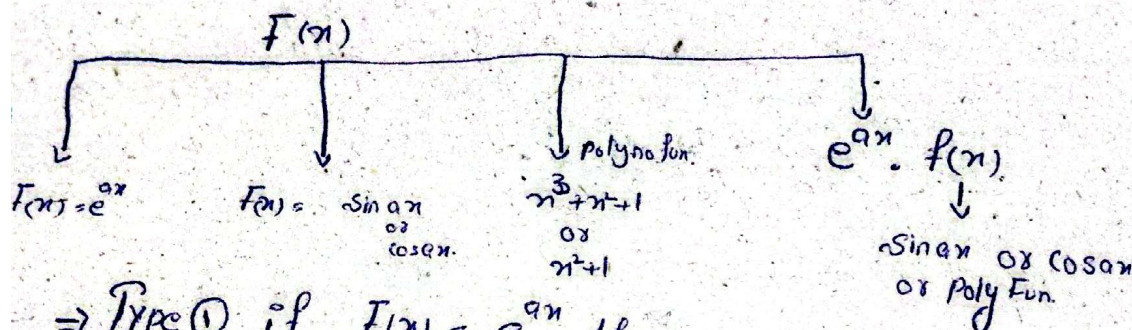


⇒ Fair copy

$$\Rightarrow y_p = \frac{1}{F(D)} F(x)$$



⇒ Type ① if $F(x) = e^{ax}$ then

$$y_p = \frac{1}{F(D)} e^{ax} \text{ then } \frac{1}{F(a)} e^{ax} \rightarrow \textcircled{1}$$

→ Note in some case $\frac{1}{0}$ occur the we will differentiate.

⇒ Type ② if $F(x) = \sin ax$ or $\cos ax$ then

$$y_p = \frac{1}{F(D)} \cdot \sin ax \text{ or } \cos ax \text{ then}$$

replace D^2 by $-a^2$

⇒ Type ③ if $F(x) = \text{Polynomial function}$ then

$$y_p = \frac{1}{F(D)} f(x) \text{ then}$$

we apply binomial series

$$(1+x)^n$$

or

$$(1+x)^{-1} \text{ then } 1 - x + x^2 - x^3 + \dots$$

$$(1-x)^{-1} \rightarrow 1 + x + x^2 + \dots$$

⇒ Type no 4

if $f(x) = e^{ax} f(x)$ $\sin x$ or $\cos x$

or poly f . then

$$\text{then } y_p = \frac{1}{f(D)} e^{ax} \cdot R(x)$$

then we use shift theorems

$$= e^{ax} \cdot \frac{1}{f(D+a)} R(x)$$

⇒ Ex. 9.2

ex. $(D^2 + 3D - 4)y = e^{2x}$

$D^2 + 3D - 4 = 0$ Auxiliary Equation $\rightarrow$ step 1

$$D^2 + 4D - D - 4 = 0$$

$$D(D+4) - 1(D+4) = 0$$

$$(D-1)(D+4) = 0$$

$$D = 1 \quad / \quad D = -4$$

Roots are real and distinct.

$$\text{Cof} = c_1 e^x + c_2 e^{-4x}$$

$\rightarrow$ step 2.

$$y = \text{Cof} + \text{PI.}$$

$$y = c_1 e^x + c_2 e^{-4x} + \frac{1}{6} e^{2x}$$

Ans.
—

$$y_p = \frac{1}{f(D)} e^{2x}$$

$$\frac{1}{D^2 + 3D - 4} e^{2x}$$

$$\frac{1}{(D^2 + 3D) + 4} e^{2x}$$

$$\frac{1}{4 + 6 + 4} e^{2x} = \frac{1}{6} e^{2x}$$

⇒ Failure case in exponential case:-

$$Q1 \quad (D^2 + 3D - 4)y = 15e^{2x}$$

$$D^2 + 3D - 4 = 0 \quad \text{Auxiliary eq.}$$

$$m = -4, 1$$

$$y_c = c_1 e^{-4x} + c_2 e^{x}$$

$$\text{Now } y_p = \frac{1}{f(D)} 15e^{2x}$$

$$\frac{1}{1+3-4} e^{2x} = \frac{1}{0} \quad \text{undefined} \rightarrow \text{Failure case}$$

so for that

$$\frac{1}{D^2 + 3D - 4}$$

$$\Rightarrow y = y_c + P.I$$

$$\times \frac{1}{2D+3-0} 15e^{2x}$$

$$y = c_1 e^{-4x} + c_2 e^x + 3xe^{2x} \quad \text{Ans}$$

$$\times \frac{1}{2D+3} 15e^{2x}$$

$$D = \frac{d}{dx}$$

$$\frac{1}{5} 15e^{2x}$$

$$3xe^{2x}$$

$$\frac{1}{0} = \text{Integral}$$

$$\frac{1}{2} \frac{1}{D} (x^2) = \frac{x^3}{3}$$

$$D(x^2) = 2x$$

$$\textcircled{2} \quad (D^2 - 3D + 2)y = e^n + e^{2n}$$

Sol

$$(m-1)(m-2) = 0$$

$$m=1 \mid m=2$$

$$\boxed{C.F = e^n + e^{2n}}$$

Now,

$$y_p = \frac{1}{f(D)} e^{2n} + e^n \quad \text{w.}$$

$$y_p = \frac{1}{D^2 - 3D + 2} (e^n + e^{2n})$$

$$y_p = \frac{1}{D^2 - 3D + 2} e^n + \frac{1}{D^2 - 3D + 2} e^{2n}$$

$$\frac{1}{1-3+2} + \frac{1}{4-6+2} = \frac{1}{0} \rightarrow \text{Failure case}$$

$\rightarrow$ Now,

$$n \frac{1}{2D-3} e^n + n \frac{1}{2D-3+6} e^{2n}$$

$$n \frac{1}{2-3} e^n + n \frac{1}{2(2)-3}$$

$$\frac{n}{-1} e^n + \frac{n}{1} e^{2n}$$

$$= -1 + 1 = 0$$

$$y_p = ne^n + ne^{2n}$$

$$\boxed{y = e^n + e^{2n} - ne^n + ne^{2n}}$$

Ans

$$Q.3 \quad (D^2 - 2D - 3)y = 2e^n - 10 \sin n$$

$$m = 3 / m = -1$$

$$C.F. = \underline{c_1 e^{3n} + c_2 e^{-n}}$$

$$\Rightarrow \frac{1}{f(D)} 2e^n - 10 \sin n.$$

$$\frac{1}{D^2 - 2D - 3} 2e^n - 10 \sin n. \quad \frac{1}{D^2 - 2D - 3}$$

$$\frac{1}{1 - 2 - 3} 2e^n - 10 \sin n. \quad \frac{1}{1 - 2 - 3}$$

$$\frac{1}{4} 2e^n - 10 \sin n. \quad \frac{1}{-1^2 - 2D - 3}$$

$$\frac{1}{4} 2e^n - 10 \sin n. \quad \frac{1}{-2D - 4}$$

$$\frac{1}{-4} 2e^n - 10 \sin n$$

$$\frac{-2e^n}{4} - \frac{1}{-2D - 4} 10 \sin n$$

$$\frac{-1e^n}{2} + \frac{1}{2(D+2)} \cdot 10 \sin n.$$

$$\frac{-e^n}{2} + \frac{(D-2)}{(D+2)(D-2)} \cdot 5 \sin n$$

$$= \frac{-1e^n}{2} + \frac{5(D-2) \sin n}{D^2 - 4}$$

$$\frac{-1e^n}{2} + \frac{5(D-2) \sin n}{-1^2 - 4}$$

$$\frac{-1e^n}{2} + \frac{5[0 \sin n - 2 \sin n]}{+5}$$

$$\frac{-1e^n}{2} + 5(\cos n - 2 \sin n)$$

$$y = y_c + y_p$$

Q4 $(D^3 - D^2 + D - 1) = 4 \sin x$ $\Rightarrow$ So Now..

Sol

$$1 - (1)^2 + 1 - 1$$

$$1 - 1 + 1 - 1$$

$$\boxed{n=1}$$

1	1	-1	1	-1
	1	1	0	1
	1	0	1	0

$$x^2 + 0x + 1$$

$$x^2 + 1 = 0$$

$$x^2 = -1$$

$$x = \pm \sqrt{-1}$$

$$\boxed{x = \pm i}$$

$$C.F = C_1 e^{ix} (b \cos x + b \sin x)$$

Now,

$$\frac{1}{D^3 - D^2 + D - 1} = 4 \sin x$$

$$\frac{1}{-i^2 D - (-i^2) + D - 1} = 4 \sin x$$

$$\frac{1}{-0 + 1 + D - 1} = 4 \sin x$$

$$\frac{1}{0} = 4 \sin x \text{ For loss case.}$$

$$x \frac{1}{3D^2 - 2D + 1} = 4 \sin x$$

$$3D^2 - 2D + 1$$

$$x \frac{1}{3(-1)^2 - 2D + 1} = 4 \sin x$$

$$3(-1)^2 - 2D + 1$$

$$x \frac{1}{-3 - 2D + 1} = 4 \sin x$$

$$-3 - 2D + 1$$

$$x \frac{1}{-2D - 2} = 4 \sin x$$

$$-2D - 2$$

$$-x \frac{1}{2(D+1)} = 4 \sin x$$

$$2(D+1)$$

$$-x \frac{1}{(D+1)} = 2 \sin x$$

$$(D+1)$$

$$-x \frac{(D-1)}{(D+1)(D-1)}$$

$$\frac{-x(D-1) = 2 \sin x}{D^2 - 1}$$

$$\frac{-x(D-1) = 2 \sin x}{-1^2 - 1}$$

$$+x(D-1) = 2 \sin x$$

$$2$$

$$\boxed{x \cos x - \sin x}$$

⇒ Algebraic cases

$$\rightarrow (D^4 - 2D^3 + D)y = x^4 - 3x + 1$$

$$c.f = c_1 e^x + c_2 e^{\left(\frac{1+\sqrt{2}}{2}\right)x} + c_3 e^{\left(\frac{1-\sqrt{2}}{2}\right)x}$$

Now y_p :

$$y_p = \frac{1}{D^4 - 2D^3 + D} (x^4 - 3x + 1)$$

$$\frac{1}{D(D^3 - 2D^2 + 1)} (x^4 - 3x + 1)$$

$$\frac{1}{D(1 - 2D^2 + D^3)} (x^4 - 3x + 1)$$

$$\frac{1}{D[1 - (2D^2 - D^3)]} (x^4 - 3x + 1)$$

$$\frac{1}{D} \left[\frac{1}{1 - (2D^2 - D^3)} \right] (x^4 - 3x + 1)$$

$$\frac{1}{D} [1 - (2D^2 - D^3)]^{-1} x^4 - 3x + 1$$

$$\frac{1}{D} \left[1 + (2D^2 - D^3) + (2D^2 - D^3)^2 + \dots \right] x^4 - 3x + 1$$

$$\frac{1}{D} [1 + 2D^2 - D^3 + 4D^4] x^4 - 3x + 1$$

$$\frac{1}{D}[(x^4+3x+1) + 2D^2(x^4+3x+1) - D^3(x^4+3x+1) + 4D^4(x^4+3x+1)]$$

Note: $(1+x)^{-1} = 1-x+x^2-x^3$

$$y = \frac{1}{D} [x^4+3x+1 + 2(12x^2) - 24x + 4(24)]$$

$$\frac{1}{D} [x^4+3x+1 + 24x^2 - 24x + 96]$$

Meany Integrate,

$$\frac{x^5}{5} + \frac{8x^3}{3} - \frac{21x^2}{2} + 97x$$

$$\frac{1}{D} \cdot \frac{1}{(D^2+1)} x^2 + \frac{x}{D} \frac{1}{3D^2+1} 3\sin x$$

→ D+D
Pcase.

Q:-2. $(D^3+D)y = x^2 + 3\sin x$

sol

$$D^3+D=0$$

$$D(D^2+1)=0$$

$$D \neq 0 \quad D^2 = -1$$

$$D = \pm \sqrt{-1}$$

$$D = 0 \pm i$$

$$C.F = C_1 e^{0x} + (C_2 \cos x + C_3 \sin x)$$

Now

$$\frac{1}{D^2 \cdot D+D} x^2 + 3\sin x$$

$$\frac{1}{D^2 \cdot D+D} x^2 + \frac{1}{D^2 \cdot D+D} 3\sin x$$

$$\frac{1}{D(D^2+1)} x^2 + \frac{1}{-1^2 \cdot D+D} 3\sin x$$

$$\frac{1}{D} (1+D^2)^{-1} x^2 + \frac{x}{D} \frac{1}{3D^2+1} 3\sin x$$

$$\frac{1}{D} (1-D^2) x^2 + \frac{x}{D} \frac{1}{3(-1^2)+1} 3\sin x$$

$$\frac{1}{D} [x^2 - D^2 x] + \frac{x}{-2} \frac{1}{3\sin x}$$

$$\frac{1}{D} [x^2 - 2(x)] + \left(-\frac{3}{2}\right) x \sin x$$

$$\frac{1}{D} [x^2 - 2] - \frac{3}{2} x \sin x$$

$$y_p = \frac{x^3}{3} - 2x - \frac{3}{2} x \sin x$$

$$y = y_c + y_p$$

$$(D^2+1)^2 = 1 - n^2 + n^2$$

Q4 Quiz Question

⇒ solve the non-homogeneous differential eqs.
subject to indicated initial conditions

$$\Rightarrow y'' + y = 4x + 10 \sin x.$$

$$D^2 y + y = 4x + 10 \sin x.$$

$$(D^2 + 1)y = 4x + 10 \sin x.$$

$$D^2 + 1 = 0 \text{ Auxiliary eq.}$$

$$D^2 = -1$$

$$D = \pm \sqrt{-1}$$

$$D = \pm i$$

so C.F.

$$C.F. = c_1 e^{0x} + (c_2 \cos x + c_3 \sin x)$$

now:

$$\frac{1}{D^2+1} 4x + \frac{1}{D^2+1} 10 \sin x.$$

$$(1+D^2)^{-1} 4x + \frac{1}{-1^2+1} 10 \sin x.$$

$$(1+D^2)^{-1} 4x + \frac{1}{-1+1} 10 \sin x \rightarrow \text{Failure case}$$

$$(1-D^2) 4x + \frac{x \cdot 1}{2D+0} 10 \sin x.$$

$$(4x-0) + \frac{x \cdot 10 \sin x \times 0 \times 1}{2D \cdot D}$$

$$4x + \frac{(x \cdot 10 \sin x) \cdot 0}{2D^2}$$

$$\frac{5}{2} (1 \sin x + \cos x) 4x - 5 \sin x - 5 \cos x.$$

$$\frac{4x + Dn \cdot 10 \sin x}{2D^2}$$